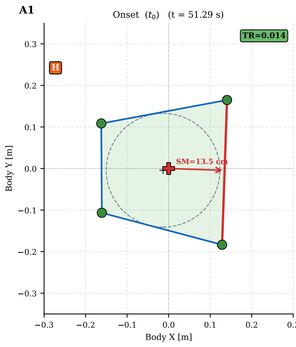
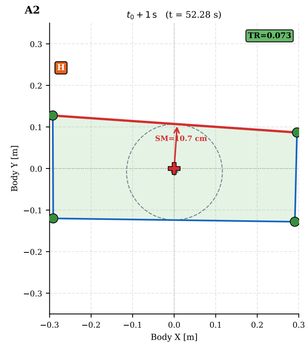
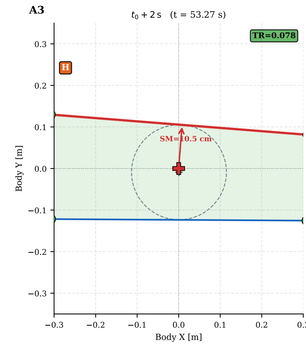
PETER Quadruped — Mode Transition Stability Analysis
Generalised McGhee-Frank (1968) Support Polygon · 4-Foot Stance

Crawl (C)

Hybrid (H)

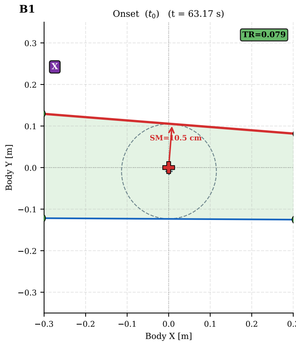


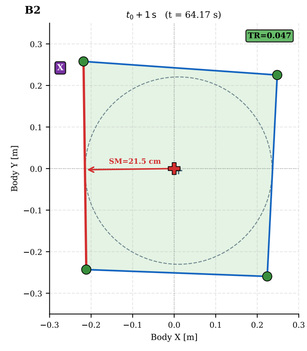


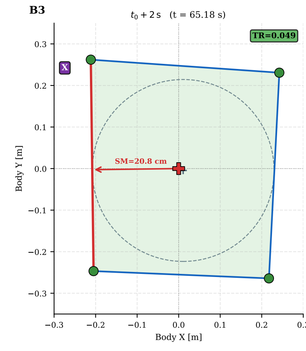


Hybrid (H)

Cross (X)

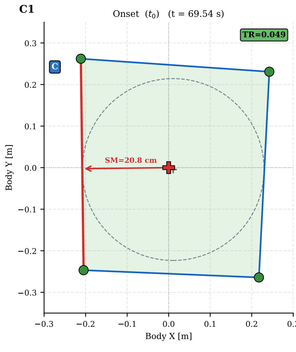


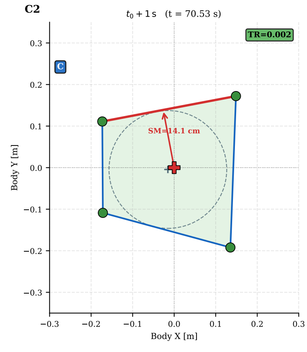


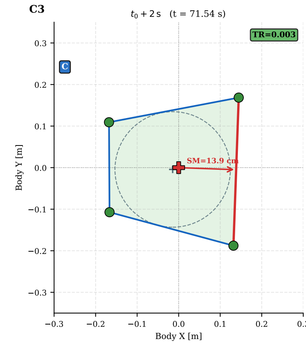


Cross (X)

Crawl (C)

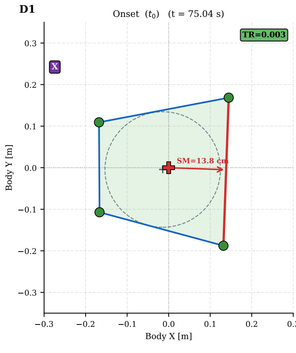


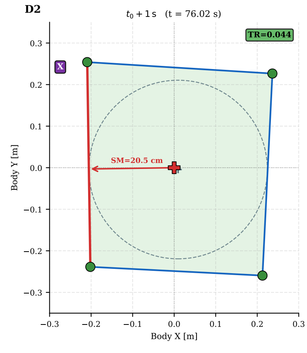


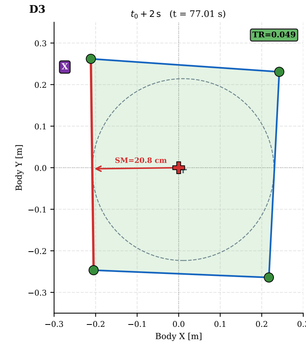


Crawl (C)

Cross (X)

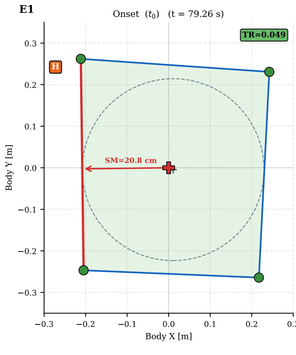


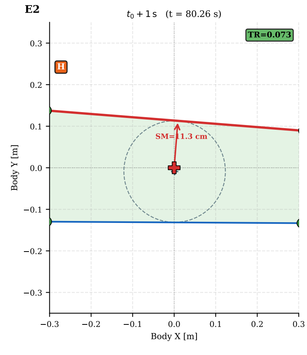


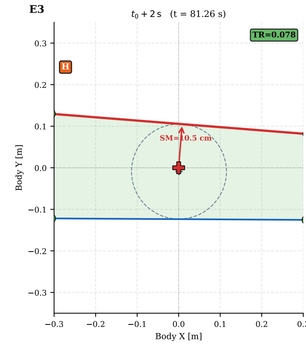


Cross (X)

Hybrid (H)

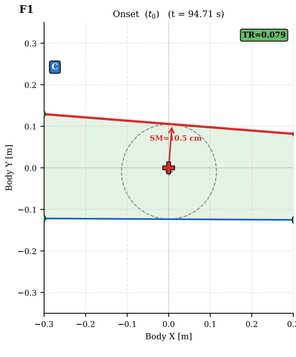


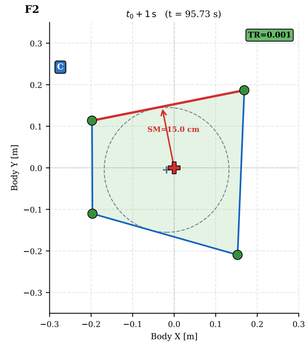


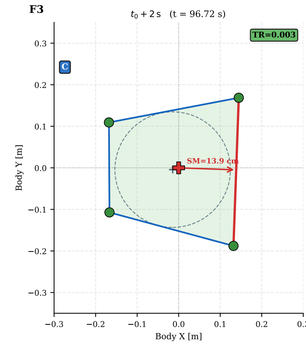


Hybrid (H)

Crawl (C)







Support Polygon
Critical Edge
Inradius Circle (r_{in})
SM Arrow (CoM → Critical Edge)
Polygon Centre
Stance Feet
Swing Foot
CoM (base_link)
Mode C — Crawl
Mode H — Hybrid
Mode X — Cross